Influence_Consensus_agr_inf - In society, at the population level, does Agriculture alone generally have a direct effect on Infrastructure, or are observed relationships primarily indirect? This is a macro level analysis, including all of society, not just agrarian or farming populations. If a direct effect exists, is it negative, none, negligible, or positive? What would be a reasonable normalized magnitude for this influence on a scale from –1.00 to 1.00? Please only provide evidence for only direct influences

Agriculture's Direct Effect on Infrastructure: Evidence Points Mainly the Other Way Around

Most available research examines how **infrastructure affects agriculture**, not how agriculture shapes infrastructure. Across these studies, infrastructure is modeled as a determinant or mediating factor for agricultural outcomes, so clear **direct causal effects of "agriculture → infrastructure" at the population level are not quantified**.

Direction of Causality in the Evidence

- Multiple macro-level studies treat **rural/transport/social infrastructure as an input** into agricultural productivity, gross value added, or sector performance, not as an outcome explained by agriculture (Zhanseitov et al., 2024; Utuk et al., 2024; Manjunath & Kannan, 2017; Djokoto et al., 2022; Preña & Tuaño, 2025).
- Infrastructure (roads, irrigation, markets, electricity, education/health facilities) is repeatedly identified as a **pre-requisite or catalyst** for agricultural growth and welfare, not vice versa (Utuk et al., 2024; Djokoto et al., 2022; Rada et al., 2020; Preña & Tuaño, 2025).
- Where bidirectional issues are discussed, concern is about **reverse causality "infrastructure ↔ agricultural growth"**, but modeling still treats infrastructure as the policy variable and agriculture as the response (Utuk et al., 2024; Preña & Tuaño, 2025).

Any Direct Agriculture → Infrastructure Effects?

- Some papers note that agriculture is a **structurally important sector** and a large user of land and rural space, contributing to the need for infrastructure, but do not estimate direct effects of agricultural output/size on infrastructure stocks or quality (Zhanseitov et al., 2024; Biaou et al., 2014).
- One study states that **agricultural production "largely determines" the condition of rural social infrastructure**, but this is a qualitative claim; no regression is presented with infrastructure as dependent and agricultural variables as predictors (Frija et al., 2020).

How Variables Are Actually Modeled

Dependent variable	Infrastructure role	Citations
Agricultural output/productivity, GVA	Key predictor (roads, irrigation, markets, electricity, composite indices)	(Utuk et al., 2024; Manjunath & Kannan, 2017; Djokoto et al., 2022; Rada et al., 2020; Preña & Tuaño, 2025)

Dependent variable	Infrastructure role	Citations
Welfare, poverty, green development	Mediator or predictor	(Kaur & Kaur, 2023; Pylypenko et al., 2022; Hua et al., 2024)
Infrastructure itself	Not modeled as a function of agriculture	(Zhanseitov et al., 2024; Utuk et al., 2024; Manjunath & Kannan, 2017; Djokoto et al., 2022; Rada et al., 2020; Preña & Tuaño, 2025)

FIGURE 1 Studies model infrastructure as cause, agriculture as outcome

Magnitude and Sign on –1.00 to 1.00 Scale

Because none of these macro-level studies estimate **infrastructure as a dependent variable explained directly by agricultural level, growth, or structure**, they do not provide a numeric elasticity or coefficient for "agriculture → infrastructure." Thus:

- **Direct effect sign:** Not empirically established; evidence is largely conceptual or qualitative.
- **Reasonable normalized magnitude:** Cannot be inferred from these models without speculation.

Conclusion

Within the provided research, observed relationships between agriculture and infrastructure are **primarily indirect or modeled in the opposite causal direction (infrastructure → agriculture)**. There is **no robust, quantified evidence** of a direct macro-level effect of "agriculture alone" on infrastructure, so assigning a normalized influence magnitude on the –1.00 to 1.00 scale is not supported by these data.

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